

Program Question_2

```
Implicit none
Integer::i,j,n
Real::x0,h
Real,Dimension(5,5)::R
x0=2.0
h=0.1
n=5
Do i=1,n
    R(i,1)=(f(x0 + h)-f(x0 - h))/(2.0*h)
    h=h/2.0
End Do
Do j=2,n
    Do i=j,n
        R(i,j)=R(i,j-1)+((R(i,j-1)-R(i-1,j-1))/(4.0**(j-1) - 1.0))
    End Do
End Do
Write(*,*)"    O(h^2)           o(h^4)           O(h^6)           O(h^8)           O(h^10)"
Write(*,*)"_____ "
Do i=1,n
    Write(*,'(5F15.10)')(R(i,j),j=1,i)
End Do
Write(*,*)"Estimated Derivative of f'(2.0)",R(n,n)

contains
Function f(x) result(f_val)
    Real,Intent(In)::x
    Real::f_val
    f_val=x*Exp(x)
End function
```

End Program